

A performance analysis of UAV landing methods

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Abstract—The following paper aims to test 3 methods (GNSS, RTK, and Optical Flow) of landing fully autonomous UAV drones, with the goal of statistically evaluating their performance. This testing was carried out using a physical FPV drone at the HC Andersen airport in Odense, with the intent of using an RTK network operated by Quadsat. However, due to time constraints with the RTK network’s availability an alternative was found using a publicly available RTK service – which had spotty coverage. Additionally, issues with the third-party Optical Flow hardware made it impossible to implement it before the scheduled test flights. Even so, 3 methods of landing were tested by substituting Optical Flow with manual landing executed by an experienced pilot.

I. INTRODUCTION

As the introduction of various drone applications is becoming more and more autonomous e.g. in logistics, medical transport, and security surveillance, it becomes increasingly interesting to investigate the performance of various methods of implementation. More specifically, this paper will focus on the performance of a subset of the entire UAV implementation; landing. Namely, the paper will focus on 3 different methods used for autonomous landing; GNSS-based landing, GNSS-RTK-based landing, and manual landing. The aforementioned methods will be tested in real-world scenarios carried out at the HC Andersen airport and Bolbro in Odense, through a long set of test flights. Once data from the various data methods have been gathered, a series of statistical analyses will be performed for the best comparison of the landing methods.

II. RELATED WORKS

Multiple studies have been carried out aiming to review different landing methods for autonomous drone operations, Long Xin et al. reviewed vision-based landing methods in static, dynamic, and complex scenarios [1]. Aiming to illustrate the advantages of Vision-based landing, as it usually comes at a lower cost and strong autonomy. However, Long Xin et al. identifies two main issues: Marker design and “the ability of anti-interference”. As such, when testing vision-based landing methods during this project these problematic attributes should be taken into account during testing and implementation.

In another work, Guatam et al. have surveyed a wide range of landing methods ranging from simple nonlinear control

methods to more advanced vision-based solutions, whilst comparing the methods in a nonstatistical manner but focusing on a design-based comparison [2]. Moreover, Guatam et al. discuss the lack of an ideal sensor for automated landing, as each sensor type comes with different issues, be it spotty GPS signals or inconsistent light levels for vision-based solutions.

III. LANDING METHODS

Before covering the testing and implementation of the various landing methods, it is important to grasp how the methods work, as this can not only provide needed insight into expected results but also enable certain precaution measures before the implementation. As such, the following subsections will loosely cover some needed theory regarding the functionality of the tested landing methods.

A. GNSS-based landing

GNSS (often referred to as GPS even though this is the American protocol) refers to a global network of navigation satellites orbiting around the earth, tasked with providing the position of users for navigational purposes.

The way this is done in practice is a bit complicated to cover here, but in simple terms, an agent can estimate its position based on the satellites it can “see”. Essentially, the GNSS client (aka rover) utilizes the distance to a minimum of 3 satellites, and via a series of calculations estimates where on the planet it is located. For a more in-depth rundown of the theory behind GNSS-based localization, we recommend reading this blog post: <https://ciechanow.ski/gps/> made by Bartosz Ciechanowski.

Although widely used, a GNSS solution usually only provides an accuracy of 30-500 cm varying on conditions such as a view of clear sky, available satellites, multipathing, and more [3]. This, however, can be improved upon by using different methods such as, but not limited to, DGPS or RTK.

DGPS, or Differential GPS, does exactly what it sounds like; it utilizes a stationary base with a known position that can be used to calculate offset to individual rovers. This provides higher accuracy, however, this can still be improved upon as covered in the next section.

B. RTK assisted landing

RTK, also known as Real-Time Kinematic positioning, is an extension of traditional GNSS solutions that provide higher accuracy through an RTK base that utilizes estimations of ionospheric effect to compute the exact satellites from which it has received positional information. This is done by decrypting a pseudorandom message sent from the GPS satellite and figuring out where it is received on the carrier wave's period. Through doing so, it is possible to obtain accuracies down to mm precision. Here, we once again refer to Ciechanowski's blog post for a deeper understanding of the loosely covered topic: <https://ciechanow.ski/gps/>.

However, as GPS RTK requires a rover to be registered with an RTK base, the availability of the solution can be limited and is often implemented using a self-operated RTK base with limited range for transmission of correction data, or through a pricey license with an RTK service covering larger areas. Even so, RTK requires a way to transmit correction from base to rover and often uses either Radio or the Internet with protocols such as NTRIP.

Lastly, it is not always certain that RTK can provide mm accuracy as it depends on the available satellites and decryption of the carrier wave. This results in RTK taking on two different states depending on this; RTK float and RTK fix. Where RTK float refers to the state in which the RTK is not properly "solved", and results in a worsened performance similar to the accuracy of DGPS.

C. Optical Flow landing

Optical flow or optic flow, is a method used to describe the apparent motion of objects in an image between two consecutive frames – as a result of either a moving camera or a moving object. The movement itself can be represented with a 2D vector field, with each vector being a point's displacement between frames. There are a multitude of methods used for implementing optical flow, but these are beyond the scope of this report. As such, this only briefly covers the idea of optical flow, but should readers be interested in investigating this topic further information regarding the topic can be found in various ways such as, but not limited to, OpenCV's documentation of the topic found here.

D. Manual landing

The Manual landing incorporates the skill of the pilot and how well the drone is stabilized. The drone was controlled in stabilized mode, where the drone actively counteracts external disturbances and pilot input to maintain a consistent level of flight (assuming that its horizontal position was calibrated perfectly). When the pilot releases control inputs, the system automatically strives to sustain the drone's roll and pitch angles close to 0 degrees (depending on how well it is calibrated - the drone we used had a bit of offset which had to be eliminated by the pilot. The throttle, in this case, was not even centred (due to personal usage of drone and controller - FPV) so there was no help in altitude. The pilot, Adam Kolodziej, was deemed a fair competitor to the autonomous landing methods.



Fig. 1. Main testing platform with H-RTK F9P and Optical Flow 3901-L0X

IV. HARDWARE

A. Main testing platform

The primary platform utilized is the FPV racing 5" drone which fully equipped can be seen in Fig. 1. This drone facilitates easy testing of various configurations. It incorporates the Mateksys H743-SLIM V3 flight controller, capable of running Ardupilot (specifically, Arducopter). For personal use, this flight controller was originally programmed with INAV, a navigation-enabled flight control software. However, the need for RTK support compelled a transition to Ardupilot. Following the flashing and configuration process, flight tests were conducted. It was discovered that Ardupilot was not optimized for overpowered builds like the FPV racing 5-inch. Consequently, multiple filters had to be set up, and PID settings tuned to achieve satisfactory flight performance. With a reasonably stable platform established, attention shifted to the critical hardware aspects of the project.

B. GNSS modules

1) *HERE+ RTK GPS*: This module is developed by CubePilot with cooperation with Ardupilot. Its target group is open source community. HERE+ GPS kit consists of a base and a rover which uses UBLOX NEO-M8P-0 module. It is easy to be set up and operated with Mission Planner, PX4 and QGroundControl. Accordingly to information from manufacturer, it was the first Differential Global Positioning System (DGPS) module for open source flight controller users. It provides RTK technology support and is equipped with a compass.

2) *Holybro H-RTK F9P Helical*: H-RTK F9P Helical can be used as a rover or base station module. It is meant for professional applications and it is also much more expensive than HERE+ which was mentioned in Point IV-B1. It provides multi-band RTK and can connect to GPS, GLONASS, Galileo and BeiDou satellite systems. It uses UBLOX F9P module and IST8310 compass. H-RTK F9P Helical is equipped with

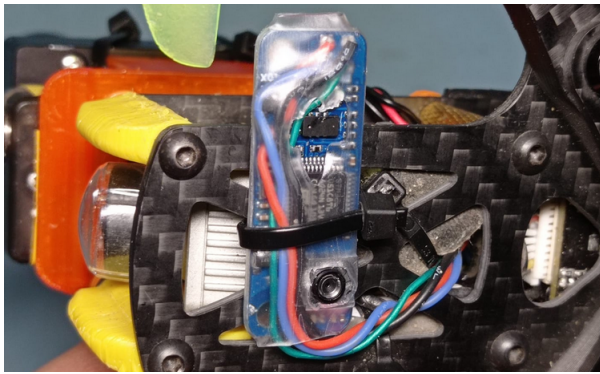


Fig. 2. Optical Flow 3901-L0X attached to UAV

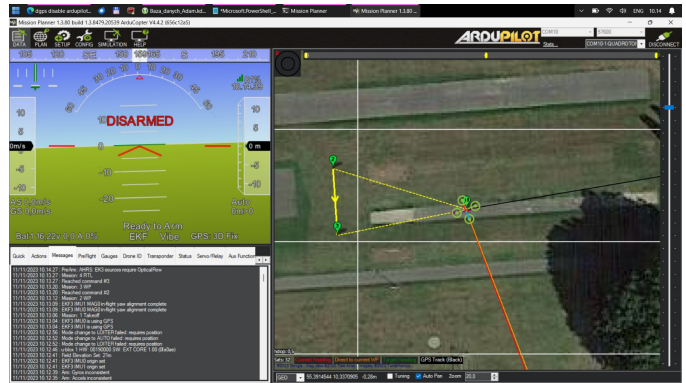


Fig. 3. Mission Planner with the automatic mission path

external helical antenna which maximizes its performance. It is compatible with the open source Pixhawk-Standard Flight Controller flight controller

C. Mateksys Optical Flow 3901-L0X

MatekSys product (Fig. 2) is a small board which includes an optical flow sensor and short-range lidar (2m). It is meant for hobby purposes and applications where small weight and size are crucial. It communicates with flight controllers via MSP protocol.

V. IMPLEMENTATION AND FLIGHTS

Automatic landings were performed based on the created mission using Mission Planner software Fig. 3. The drone was programmed to take off and go up 5m altitude. Then it was supposed to fly through two waypoints and perform RTL (Return To Launch). Thirty kinds of missions were performed for each GNSS-based and RTK-based test. The last series of data was gained based on manual landings in Stabilized mode. This section covers the process of research flights and faced problems.

Video with flight examples can be found under the following link:

Scientific Methods - flights examples

The First flight shown at the beginning of the video was done at the HCA Airport with HERE+ module. Data collected during this flight wasn't used in the final database. The rest of the video shows correct flights which were part of the research.

A. First research flights and problems with HERE+

Although the drone was originally equipped with Mateksys M8Q-5883 GPS & Compass module had to be replaced with a module which would be able to utilize RTK corrections data. The HERE+ RTK GPS was provided by SDU (with special thanks to Frederik Mazur Andersen and Oscar Bowen Schofield). More information about this module can be found in Hardware section IV-B1. Ardupilot was configured to exchange data with this GPS and flights based on satellites were successfully performed. The intention was to use Quadsat's



Fig. 4. UAV with HERE+ and Optical Flow 3901-L0X during one of the flights at the HCA airport

RTK (special thanks to the company) which was supposed to work through TCP Client framework. Following the standard procedure for connecting RTK to Quadsat's drones, a laptop interfaced with the WiFi network to transmit RTK corrections from the base station to the drone via an integrated telemetry link on our platform. Unfortunately, the envisioned setup did not yield the anticipated results. Despite multiple attempts, after a few conversations with more experienced people, no discernible progress was achieved. Ultimately, the conclusion was drawn that the HERE+ RTK GPS module was likely faulty.

B. GNSS module replacement and RTK-based research flights

Luckily new module - Holybro H-RTK F9P Helical IV-B2 was received as a replacement for HERE+. This module is much newer and much more advanced compared to HERE+. It automatically switches to 3D DGPS mode (more information III-A). Due to lack of time, flights couldn't be performed at the HC Andersen airport during its working hours. That meant that the Quadsat's RTK base also couldn't be used. It was required to switch to NTRIP source (more information III-B). The free provider was chosen - Point one Navigation.

Unfortunately, the closest RTK base was in Aarup. It turned out that it was probably too far to get corrections which would result in an RTK fix. However, it allowed us to get RTK Float. This experiment didn't require infinite precision so it wasn't a huge problem.

C. Optical Flow calibration and final series of data

The next series of data was supposed to be collected using the Optical Flow Sensor explained in section IV-C. The sensor was connected is set according to manuals provided by the manufacturer and Ardupilot documentation. Calibration had to be performed. The process started correctly but the problem was that it could not be finished. Several tries were made. The drone was flown 1,5m above the ground because the sensor has a maximum of 2m of range. Every try ended up with a "timeout" message about 70% of calibration finished. It was probably the problem with the board which is now being solved with the manufacturer.

To fulfil requirements, a third series of data had to be collected. How it works can be read in the section III-D. After every takeoff, a pilot performed some movements with the drone and then came back to the landing spot. It was done 30 times - the same number as the rest of the test.

VI. RESULTS

Each landing method was carried out 30 times to reach a good distribution of data. The landing was measured as an error according to the starting position. All the data can be seen to be normally distributed (see fig5). It can be seen from the box plot (see fig 6) that there are a few outliers, for looking for statistical significance ANOVA was performed in table I. From there, it can be seen that there is a statistical significance.

TABLE I
ANOVA

source	ss	df	ms	f	p
data	15527.6	2	7763.81	31.14	6.32073e-11
error	21632.9	87	249.34		
total	37220.5	89			

To find the significance, a T-test with Bonferroni correction was done in table II, from which it can be seen that GPS and RTK are not statistically significant and manual is statistically significant from the rest then using an α of 0.05.

TABLE II
T-TEST

	GPS vs RTK	RTK vs Manual	GPS vs Manual
T-Test	0.4969	8.786e-07	5.3902e-10
bonferroni	0.999	2.6358e-06	1.6171e-09

VII. CONCLUSION

This paper aims to test 3 different autonomous landings, GNSS, RTK, and Optical flow. Due to hardware problems, Optical flow did not work, and RTK was used with an online RTK service, with a poor signal. The scope of the project of

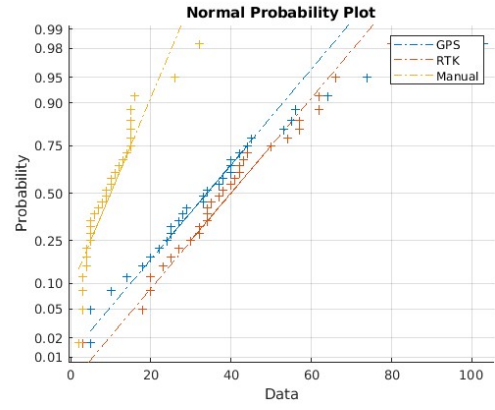


Fig. 5. normal distribution check

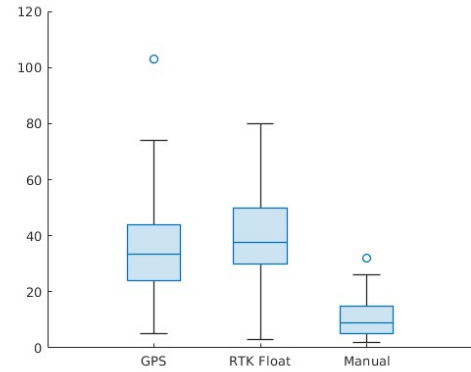


Fig. 6. The boxplot shows the quantiles of GPS, RTK, and Manual landing

the project was then changed. instead of landing with optical flow, an experienced pilot would manually land the drone.

The hypothesis, that the RTK landing would be the most accurate, can therefore not be rejected or accepted, because an invalid RTK fix was recorded.

It can be concluded that the manual landing performed better than the autonomous landings since the normal distribution and mean are lower. This shift in methodology highlights the importance of adaptability in drone operations and the need for further research to enhance the reliability of autonomous landing systems.

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